

Institute of Actuaries of India

Subject CP1 – Actuarial Practice (Paper A)

September 2021 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**i) Difficulties with data:**

The data supplied by different companies may not be comparable:

- differences in cover (Lien period, exclusions, , policy conditions)
- differences in underwriting practice
- differences in claim settlement practice
- differences in nature of data stored by different companies
- different coding used for risk factors/rating factors
- Industry-wide data may not even exist for this as not many companies have yet introduced this kind of product
- Credible Data may not be detailed enough for pricing
- Data may be out-of-date (takes time to collect, collate & distribute) due to the different variants of Covid-19
- Data quality may be poor (depends on quality of data of all contributors) as even though the deaths happened due to Covid-19 but many of them were not admitted to Hospitals and declared normal death
- Not all companies contribute so may be unrepresentative
- different geographical section of market
- different socio-economic section of market

(4)

ii) Data: Policy

- Postcode
- Age of policyholder
- Gender
- Policy number
- Premium amount
- Sum Assured
- Death Benefit (if different from Sum Assured)
- Date of Commencement of Risk Payment frequency (annual/monthly)
- Payment method (DD/credit card)
- Unique link between policy and claim, for matching
- Unique link between this policy and other products e.g. customer ID

Claims

- Incident date, reported date, settlement date
- Claim number
- Cause of death
- Paid date for all claims payments
- Claim amount paid,
- Claims handling expenses
- Claim status
- Distribution Channel (e.g. Bank, Broker etc.)
- Name of the Distribution Channel

Solution 2:

The company could provide Health Insurance to only some of its employees. This would cut the costs dramatically but would annoy those who no longer received the benefit, with the risk that they might look for another job.

The company could offer an increase in pay in lieu of providing the health insurance or ask employees to contribute towards the cost.

The company could seek to reduce the level of cover provided under its Health Insurance Policy

- e.g. by increasing the excess level
- or by capping claims
- or reducing choice of treatment providers (or other suitable examples)
- or removing some of the areas covered
- or withdrawing cover for family members/dependents
- Exclusion of maternity cover, if any, as this has substantial cost
- This would have a smaller impact on the cost but might be more acceptable to the workforce.

The risk would be that the staff might not fully understand the change.

The company could move to a budget version of cover such as major medical expenses or a health cash plan version or an optical/dental plan or a waiting list plan. This would have a greater impact on the level of costs than simply reducing the cover and would be perceived as fair if the same change was made for everyone. However, it is a noticeably reduced benefit for staff, and carries some risk that staff may leave.

The company should certainly shop around to see if it can get a better deal and should also investigate whether there are any product variants that might be advantageous, such as profit sharing.

The underwriting procedure could be tightened up e.g. have full medical underwriting rather than giving Free Cover (FCL) to their employees.

It is possible (but unlikely) that if the company is very large it might be feasible to self-insure and still provide the staff with the underlying benefits. This increases the level of risk to the company. This would require more internal administration but reduces the transfer of profit to the insurer and loses any bulk purchasing power which insurers have with healthcare providers.

The company could try to encourage better lifestyle behaviour in its employees but this would be unlikely to have an immediate effect on the cost of claims.

[9 Marks]**Solution 3:**

- i) Value at Risk generalises the likelihood of underperforming by providing a statistical measure of downside risk.

VaR assesses the potential losses on a portfolio over a given future time period with a given confidence level.

VaR is Measured either in absolute terms or relative to a benchmark
VaR is based on assumptions that may not be immediately apparent.
VaR is Frequently calculated assuming a normal distribution of returns.

(5)

- ii) It is based on assumptions that may not be immediately apparent In particular, it is frequently calculated assuming a normal distribution of returns.

If the distribution of returns is “fat tailed” or skewed, tracking error may be misleading

It doesn’t consider the outcomes within the tail

Unfortunately, portfolios exposed to credit risk, systematic bias or derivatives may exhibit non-normal distributions.

The usefulness of VaR in these situations depends on modelling skewed or fat-tailed distributions of returns either in the form of statistical distributions or via Monte Carlo simulations

However the further one gets out into the “tails” of the distributions, the more lacking the data and hence the more arbitrary the choice of the underlying probability becomes

(5)

[10 Marks]**Solution 4:**

Tom, Dick and Harry have conducted an initial appraisal relating to opening one shop.

They will have investigated the risks involved in the project. These will include:

- Finding suitable premises, staff and suppliers. (1)
- Attracting enough customers (spending enough money, as antique gifts are normally expensive). (1)
- Having sufficient capital available to fund the project. (2)

They will have come to a view on the best course of risk mitigation, having regard to the costs involved. (1)

They will need to evaluate the most likely cashflows for capital expenditure (e.g. buying/leasing shop and fitting), running costs (e.g. cost of gifts and other supplies, cost of staff) and revenues (mainly sales of gifts). Many assumptions will be needed. [4]

They can then use these cashflows to calculate the net present value of the project using an appropriate risk discount rate. They will have decided on a suitable discount rate; this may relate to rate of return that they could earn elsewhere with an adjustment for the risk involved. (2)

The cashflows would also be used to calculate the payback period. (1)

Opening a chain of shops would involve a similar process but the values and risks will change. (2)

Finding suitable premises may be more difficult as they may be less likely to have local knowledge. Similarly for suppliers. They may have to rely on the advice of others. (3)

They will have to select the premises to keep their customers in mind as there are selective customers of antique things.

There may be more competition in different locations. However, having more shops will give some diversification. (2)

They will be less likely to be personally involved and so will need to hire a manager for each shop and will have less control over other staff.

[Max 12 Marks]

Solution 5:

- i) Risk budgeting is the process of establishing how much risk should be taken and where it is most effective to take the risk (in order to maximise return).

It can be done at an organisational level i.e. the whole company, or down to a portfolio level e.g. Guaranteed return policies vs Unit Linked Policies etc. (2)

- ii) The risk budgeting process for investment risks has two parts:

- The first part is deciding how to allocate the maximum permitted overall risk between total fund active risk and strategic risk.
- The second part is allocating the total fund active risk budgets across the component portfolios.
- The key focus when setting the strategic asset allocation is the risk tolerance of the stakeholders in the fund. This is the systematic risk they are prepared to take on in the attempt to enhance long-term returns.
- The key question on active risk is whether it is believed that active management generates positive excess returns.
- Risk budgeting is, therefore, an investment style where asset allocations are based on an asset's contribution to the portfolio as well as on the asset's expected return.
- A risk budgeting strategy can free the manager to look for alternative investments that might increase the expected return on the portfolio.
- The constraint is that the total risk of the portfolio must stay at or below a target level, so increased attention is paid to low correlation investments.
- Allocations to such investments can reduce the total risks of the portfolio through diversification.

(7)

[9 Marks]

Solution 6:

- i) **Main Investment objective is to maximize return**

- while meeting all contractual obligations (0.5)
- Maintaining solvency (0.5)

Characteristics of assets should match with those of liabilities by term, amount, nature and currencies of liabilities (0.5 for each*4)

[3]

ii)

- Equity exposure of 35% is very high for company which write predominately short tail line of business. It is advisable to have not more than 15-20% equity exposure for general insurance company predominately writing short tail.
- Equities are better matched for long tail liabilities such as TP. Concern as most business is short tail
- Equity is less secure in comparison to Govt. Bond.
- Greater Volatility of returns so risky investment type
- Solvency is depended on level of stock market
- Credit rating agencies may not look favorably
- Board of company may object high level exposure in Equity
- Regulatory restriction on maximum percentage which can be held on equities and consideration of value admissible for solvency
- Liquidity risk as when large payments are required in short notice as pandemics or catastrophe claims
- Market risk as Future return from stock may be lower
- High level of expertise is required to buy and sell which may not be available in company

(1 Mark each)

[Max 5]

iii) **Reasons which individually or combinedly impacted solvency may be:**

- Earlier Motor TP claim provision may be set on lower side. It has been recently revised to appropriate level.
- Focus only on top Line not on Bottom line
- Increase in Inadmissible assets such as Coinsurance recovery, reinsurance recovery, due from brokers and policyholders
- New recent judgments in Motor TP which has increased liability
- Fall in assets value such as equity due to low level of Market
- Poor business underwritten
- Insufficient premium due to competitive pressure
- Product were filed long back and recent premium rate insufficient
- Pandemic has increased sudden outgo in Health
- One time increase in provision such as pension
- Increase in Expenses
- High default of assets and bad debts
- Change in way solvency is calculated such as introduction of Solvency II
- Increase in business which are more capital intensive
- Poor reinsurance arrangement so lesser recovery
- Soft market

(0.5 Mark each)

[Max 5]

iv)

- Company is not likely to have large amounts of free assets as solvency is low.
- Therefore, it is not likely to be a great deal of freedom to invest
- There will be some regulatory restriction on investment of a company in run-off

- There will be no new premium income except only some premium income in the short term from business written in instalment premium.
- Motor OD and Health business will runoff quickly except mainly Motor TP claims
- The company may have exposure to property catastrophes, for which a degree of liquidity is required.
- There will be greater need for liquidity which need more cash
- Investment needs to be done at a shorter term than matching would imply to counter liquidity concern.
- Government securities are better options for liquidity.
- Equities are highly volatile and have substantial market risk.
- Company has substantial equity holdings and reduction will be required.
- Equity may be replaced with high-grade corporate bonds for better return.
- The company will need to make a ALM projection and invest in assets which are greater than liability for initial durations.
- This matching will need to be done by term which may be monthly for initial months
- Generally average duration of asset is higher than duration of liabilities to maximise investment return for going concern scenarios
- Now with investment in assets with earlier maturities that would match the projected outgo, duration of assets will reduce.
- Fixed Expenses will be more substantial which increases the need for liquidity.
- Company needs to follow rigorously for agent, coinsurance and reinsurance recovery, if any
- It needs to look for prospects to commute its liabilities favourably to quickly run-off.
- Different line of business is affected by a number of different inflation rates.
- Liability and latent claims may take longer to settle which need hedge against inflation.
- Tax implications will need to be considered

(0.5 Mark Each)

[Max 7]

[20 Marks]**Solution 7:**

i)

- virus damage
- Forensic analysis for identifying the attack source
- Public relations services
- Notification of clients
- Reputational damage
- Cost of restoring/repairing/replacing damaged IT systems
- Payment of regulatory fines
- hacker attack
- Social Engineering
- infringement of intellectual property
- Putting in place measures to protect any affected customers
- slander or libel
- stalking or bullying
- Identity theft
- personal data or privacy issues (e.g. loss of or disclosure of private data)
- ransomware / malware
- business interruption/ Loss of income costs to help restore business functions and to maintain

business capabilities while responding to a cyber incident

- losses relating to breaches or actions by suppliers or business partners
- Investigation, defence costs and civil damages associated with security and privacy breaches of customer data
- forensic investigation costs to determine the source of the cyber incident/ breach and the extent of harm caused
- remediation costs to rectify any network problem or software deficiencies
- notification costs to customers whose data was compromised
- data restoration costs of data stolen, lost, or altered
- legal costs to evaluate regulatory obligations and assess any liability

(0.5 Mark Each)

[Max 5]

ii)

Policy Terms:

Sum Insured

Legal costs

Excess and limits

Loss “occurrence” basis, or claims made basis,

(0.5 Mark Each)

Size and Industry:

Revenue/turnover

Type of Business

Area of operation of Business

Number of employees

Have employees been properly trained on how to mitigate risk?

If the company allows employees to use personal devices or work remotely

Past experience

Business continuity plan in case of IT event / backup of servers etc

Dependence of the revenue on IT

(1 Mark Each)

IT system:

- Firewalls in place or anti-virus software in place
- password policies, frequency of patches and other updates
- data encryption policies
- The third parties’ data/IT security standards with whom the company shares data with or receives data
- Level of network access provided to third-party vendors
- Monitoring of vendors who has access to computer and data
- Location of servers e.g. cloud or particular host

(1 Mark Each)

Data:

Volume of data the company holds

Complexity of data/systems architecture

Sensitivity of data captured/held

(1 Mark Each)

Others:

Trends in incidence and severity of cyber attacks

Regulatory fines

(0.5 Mark Each)

[Max 7]

iii)

a) Opportunities

- Diversification
- Exposure to an expanding market
- Elevating company standing
- Access to potentially profitable business
- Might cross sell other policies

(0.5 Mark Each)

[Max 2]

b) Pricing challenges and other risk in pricing the cover:

- Lack of knowledge of the market
- Product design may not be appropriate
- Potentially unprofitable if market soft
- Low number of high value policies – high variability of claims/profit
- May be capital intensive even with diversification
- Unavailability of data
- difficulty in getting reinsurers' support or available at high rates.
- Another country data may be of limited use due to different legal environment and coverages
- difficulty in assessing the severity
- The inflation and other trends are difficult to be determined.
- Specialised skill may be required in underwriting and pricing.
- Lack of historical events to estimate likelihood of losses
- Chances of accumulation, e.g. a worldwide ransomware/hacking attack affecting many insured
- Delay in reporting as claim has been noticed later
- Immense ways in which security could be breached, and
- Ways in which security can be breached is evolving and not known at start of cover
- Developing regulatory requirements
- Unpredictability about fines for breaches
- High level of moral hazard or anti-selection
- Difficult to ascertain future business resulting in improper loading for fixed Expenses

(0.5 Mark Each)

[Max 4]

c) Ways to Manage above risk:

- Reinsurers' may provide support in pricing and claim settlement
- Company can start with coinsurance and slowly develop data and expertise
- Industry data, if any available could be used.
- Capping of severity amount, instead of offering unlimited liability for the policy.
- Frequency of claims may be estimated based on business, geography if available based on other data sources
- The specialised underwriters and claims personnel support may be taken
- Use excesses to avoid small claims
- Use limits to manage accumulations
- Use policy exclusions or restrictive policy language to limit risk exposure

(1 Mark Each)

[Max 4]

[22 Marks]

Solution 8:

- i) Insurer can compare its own experience with that of the industry
- at the overall level and
 - at the level of the categories into which the data is classified
 - Helps in understanding insurer if its experience is in line with Industry. Are there valid reasons for differences, if any noticed or course correction is required?
 - Industry-wide data provide a benchmark for insurers to assess their position compared to their competitors.
 - Can help identify age bands where their premium rates or claims outgo are higher or lower when compared to the competition
 - Can help insurers with low business volumes or new Insurer in benchmarking their rates
- (1 Mark Each)
[Max 4]
- ii)
- Extent of add-ons also influences the claims outgo. (1)
 - Disease details are not collected properly which heavily influence the risk (1)
 - Data quality could also be an issue (0.5)
 - Claim cost varies by cities and Hospital (1)
 - Companies will have different practices, for example, medical underwriting (1)
 - claim settlement and outstanding claim reserving policies may be outsourced to TPAs in a few companies (1)
 - Product structure and benefits are different (0.5)
 - The nature of the data stored by different companies will not always be the same (0.5)
 - The coding used for the risk factors may vary from company to company. (0.5)
 - The data will be much less detailed and less flexible than those available internally, hence it will be more difficult to manipulate. (0.5)
 - External data is often much more out of date than internal data. The data takes quite a while to collect, collate and then distribute to the insurers. (0.5)
 - The data quality will depend on the quality of the data systems of all its contributors. If one company makes a mistake (eg entering a figure in millions instead of thousands) then this invalidates the whole set of data. The more companies that contribute, the more likely that one will make a mistake! (0.5)
 - Not all companies contribute. (0.5)
- [Max 6]
[10 Marks]
